



On the resolution function for powder diffraction with area detectors

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In a powder diffraction experiment the resolution function defines the instrumental contribution to the peak widths as a function of the Bragg angle. The Caglioti formula is frequently applied to model the instrumental broadening and used in structural refinement. The parameters in the Caglioti formula are linked to physically meaningful parameters for most diffraction geometries. However, this link is lost for the now very popular powder diffraction geometry using large 2D area detectors. Here we suggest a new physical model for the instrumental broadening specifically developed for powder diffraction data measured with large 2D area detectors. The model is verified using data from two synchrotron diffraction beamlines with the Pilatus2M and MAR345 detectors. Finally, a functional form is proposed to replace the Caglioti formula for this geometry in the Rietveld method and profile refinements.

1. Introduction

The instrumental resolution $R(\mathbf{Q})$ contributes to the profile of a diffraction line $L(\mathbf{Q})$ via a convolution with the scattering function of a sample $S(\mathbf{Q})$, where $\mathbf{Q}$ is the scattering vector satisfying the Bragg condition (Klug & Alexander, 1974):

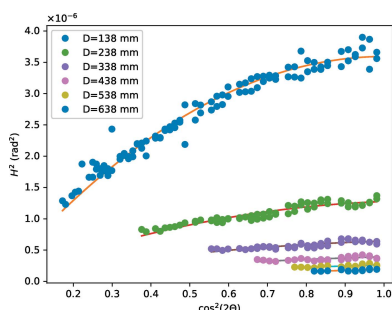
$$L(\mathbf{Q}) = S(\mathbf{Q}) * R(\mathbf{Q}). \quad (1)$$

Knowledge of the resolution function is a necessary prerequisite for quantitative analysis of the scattering function. Typical examples of quantitative applications are refinement of crystal-structure models based on intensities, and microstructural characterization based on diffraction profiles. In particular, a good model of the $\mathbf{Q}$ -dependent broadening of powder diffraction lines is crucial for the intensity calculations in the Rietveld and pair distribution function (PDF) methods (Pecharsky & Zavalij, 2009; Qiu *et al.*, 2004), especially if the experimental pattern covers a large $\mathbf{Q}$ range with many overlapping lines.

One common line-broadening function stems from early Caglioti papers (Caglioti *et al.*, 1958). Recently this was revisited specifically for a synchrotron powder diffractometer equipped with a line detector and focusing optics (Gozzo *et al.*, 2006). In a generic form, it assumes that the full width at half-maximum (FWHM, H) depends on the Bragg angle θ as

$$H^2 = U \tan^2 \theta + V \tan \theta + W, \quad (2)$$

where U , V and W can be expressed via physically meaningful parameters of a diffractometer [see Gozzo *et al.* (2006) for an example].



ometers with $\delta d/d < 10^{-3}$ are considered to be high-resolution instruments.

If we take into account that $\delta d/d = -\delta\theta \cot(\theta)$, the uncertainty for the Bragg angle is given by

$$(\delta\theta)^2 = \frac{1}{4} \tan^2 2\theta \cos^4 2\theta \left[\left(\frac{\delta D}{D} \right)^2 + \left(\frac{\delta R}{R} \right)^2 \right] + \tan^2 \theta \left(\frac{\delta\lambda}{\lambda} \right)^2. \quad (8)$$

Neglecting the uncertainty for the wavelength, one arrives at following form:

$$(\delta\theta)^2 = \frac{\cos^4 2\theta}{4D^2} [\tan^2 2\theta (\delta D)^2 + (\delta R)^2]. \quad (9)$$

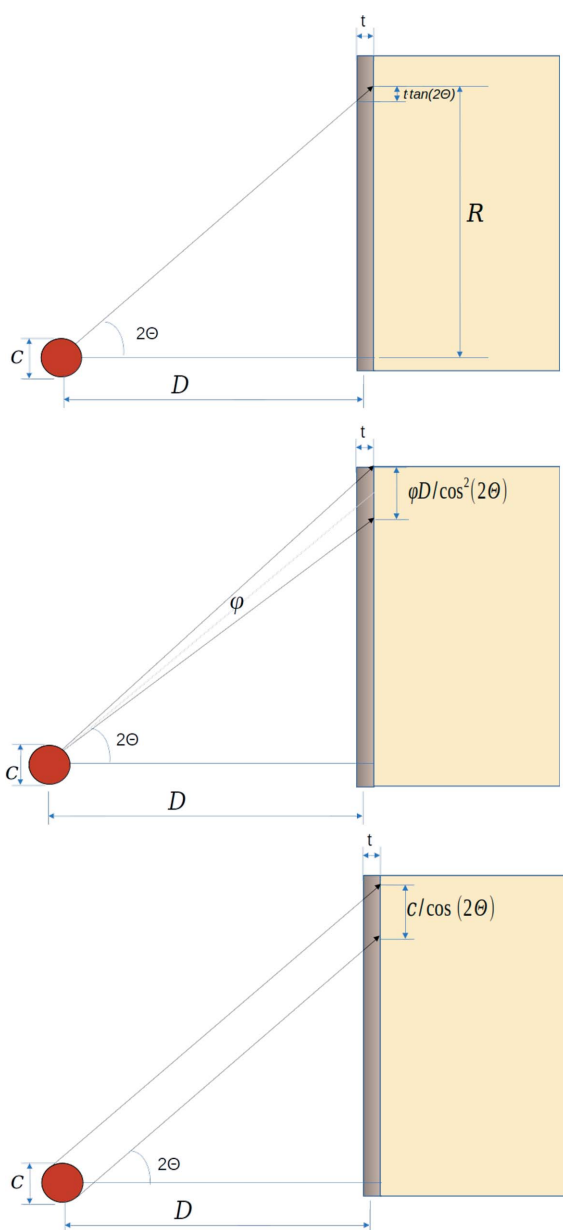


Figure 2
Contributions to line broadening associated with the thickness of sensitive layer, the beam divergence and the sample size.

Let us now enumerate the contributions to δD and δR . δD is defined by the uncertainty in the longitudinal position of the scattering event given by the size of the sample c , and by the uncertainty in the longitudinal position of detection that is less than or equal to the thickness of the sensitive detecting layer of the detector t . δR is defined by the size of the pixel or point spread function p , the projection of the sample onto the detector plane, the uncertainty in the transverse position of detection associated with the thickness of the sensitive layer t , and the divergence of the scattered beam ϕ . All of these contributions are shown schematically in Fig. 2.

Finally, assuming normal distributions for all the components and taking into account error propagation (Appendix A), the uncertainty for the Bragg angle reads

$$(\delta\theta)^2 = \frac{\cos^4 2\theta}{16D^2} \left[\tan^2 2\theta (c^2 + 2t^2) + p^2 + \frac{c^2}{\cos^2 2\theta} + \frac{D^2 \phi^2}{\cos^4 2\theta} \right]. \quad (10)$$

Taking the angular uncertainty as a variance for the normal distribution, the FWHM can be written as

$$H = 2(2 \ln 2)^{1/2} \delta 2\theta = 4(2 \ln 2)^{1/2} \delta\theta, \\ H^2 = 32 \ln 2 (\delta\theta)^2 = A \cos^4 2\theta + B \cos^2 2\theta + C. \quad (11)$$

The coefficients in equation (11) are defined by the experimental conditions as follows:

$$A = \frac{2 \ln 2}{D^2} (p^2 - 2t^2 - c^2), \\ B = \frac{2 \ln 2}{D^2} (2t^2 + 2c^2), \\ C = 2\phi^2 \ln 2. \quad (12)$$

Equation (11) gives the desired functional form of the instrumental broadening for powder diffraction data collected with an area detector that is set normal to the beam (Fig. 1). In spite of its different form, this is similar to the expression for the instrumental line broadening given by equation (4) when $t = \phi = 0$.

If the detector is tilted by an angle α towards the sample, equation (5) reads

$$d = \frac{\lambda}{2 \sin \left\{ \frac{1}{2} [\arctan(R_\alpha/D_\alpha) + \alpha] \right\}}, \quad (13)$$

where R_α and D_α are defined by the point of normal incidence (PONI; see Ashiotis *et al.*, 2015). Obviously, this does not change the derivatives in equation (6) and the resolution function is therefore given by

$$\left(\frac{\delta d}{d} \right)^2 = \frac{1}{4} \cot^2 \theta \cos^2(2\theta - \alpha) \tan^2(2\theta - \alpha) \\ \times \left[\left(\frac{\delta D}{D_\alpha} \right)^2 + \left(\frac{\delta R}{R_\alpha} \right)^2 \right] + \left(\frac{\delta\lambda}{\lambda} \right)^2 \quad (14)$$

and, as before, neglecting the uncertainty of wavelength gives the angular uncertainty as

$$(\delta\theta)^2 = \frac{\cos^4(2\theta - \alpha)}{4D_\alpha^2} [\tan^2(2\theta - \alpha)\delta D_\alpha^2 + \delta R_\alpha^2]. \quad (15)$$

Equation (11) is therefore also applicable to data collected with a tilted detector by simply replacing 2θ with $2\theta - \alpha$.

For a high-resolution regime, one that assumes long sample-to-detector distances D and small values of coefficients A and B in equation (12), the effect of the uncertainty in the wavelength, $\delta\lambda/\lambda$, has to be considered explicitly. The instrumental broadening can then be written as

$$H^2 = Ax^4 + Bx^2 + C + M \frac{1-x}{1+x}, \quad (16)$$

where $x = \cos(2\theta)$ or $x = \cos(2\theta - \alpha)$ and $M = [4(2\ln 2)^{1/2}(\delta\lambda/\lambda)]^2$. Several graphical representations of equation (16) are given in Appendix A for a range of values of the parameters. It is clear that for all the considered cases the Caglioti formula cannot deliver the parameterization of the instrumental broadening as modelled with equation (11). In the next section we check whether equation (11) meets reality.

3. Instrumental resolution – experimental verification

In order to illustrate the validity of the theoretical derivations above, we have measured a set of powder diffraction patterns at different sample-to-detector distances. These data were collected at the BM01 station of the Swiss–Norwegian Beamlines, SNBL, at the European Synchrotron Radiation Facility, ESRF (Grenoble, France), using the Pilatus@SNBL diffractometer equipped with a Pilatus2M area detector (Dyadkin *et al.*, 2016). The sample, LaB₆ NIST Standard Reference Material N660b, was measured with 0.57098 Å (21.71 keV) radiation at sample-to-detector distances of 138, 238, 338, 438, 538 and 638 mm and vertical offsets of 0 and 80 mm to extend the range of Bragg angles covered. The second set of measurements were done with a constant sample-to-detector distance, 138 mm, but at different energies

of the synchrotron radiation in the range 15–20.5 keV. The measured patterns were used to calibrate the diffractometer parameters with *PyFAI* (Kieffer & Karkoulis, 2013) and the 2D data were azimuthally integrated to 1D powder diffractograms with *Bubble* (Dyadkin *et al.*, 2016). All line positions, intensities and profile parameters were refined using a pseudo-Voigt profile function.

We have fitted equation (11) to the experimental FWHMs (Fig. 3), and the model satisfactorily reproduces the experimentally observed line broadening.

Assuming the pixel size for the Pilatus2M is fixed to be $p = 0.172$ mm, the apparent sample size and the effective detector thickness can be estimated from the coefficients A and B at different sample-to-detector distances. First we note that

$$A + B = \frac{2\ln 2}{D^2}(p^2 + c^2), \quad B + 2A = \frac{2\ln 2}{D^2}(2p^2 - 2t^2) \quad (17)$$

are linear functions of $(2\ln 2)/(D^2)$. The sample size and effective thickness of the detector absorption layer can therefore be estimated from the gradient of the corresponding linear fits in Fig. 4(a). The sample size was estimated to be 0.142 (2) mm; this is in between the size of the capillary (0.1 mm) and the beam size of approximately 0.2 mm. The effective thickness was estimated to be 0.177 (7) mm, which is substantially lower than the nominal thickness of the silicon layer (0.45 mm). Since the penetration depth and absorption length depend on the energy of the radiation, the effective thickness of the sensitive layer also depends on the energy. The experimental dependence is given in Fig. 4(b), based on the analysis of data measured at different energies. The effective thickness is therefore less than or equal to the nominal thickness and, for a sufficiently thick layer, will increase with X-ray energy.

An additional experiment was performed at the BM31 station of the SNBL at the ESRF (Grenoble, France). BM31 is equipped with a Mar345 image plate, which is used for both

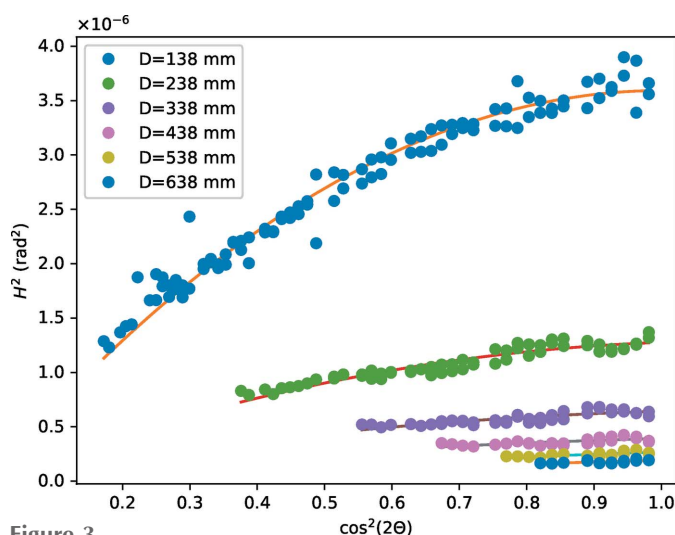


Figure 3
Resolution functions and their fits for various distances for LaB₆ measured at the SNBL BM01@ESRF beamline.

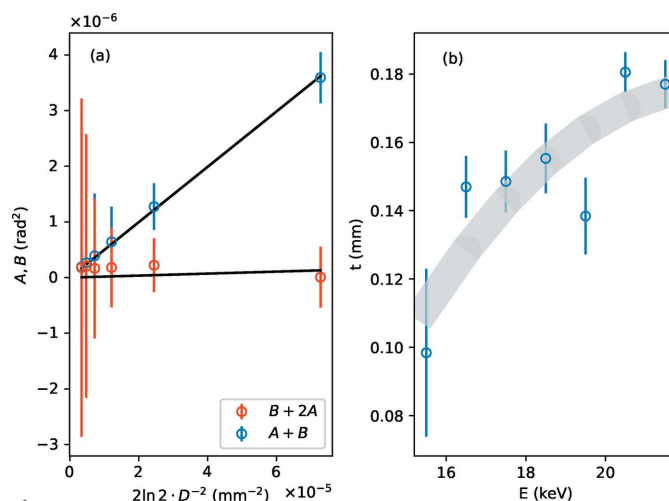


Figure 4
(a) $A + B$ and $B + 2A$ as functions of $(2\ln 2)/(D^2)$ for LaB₆ patterns from Fig. 3. Straight black lines show the linear fits. (b) Energy dependence of the effective thickness of the detector absorption layer. The region shaded grey is a guide for the eye.

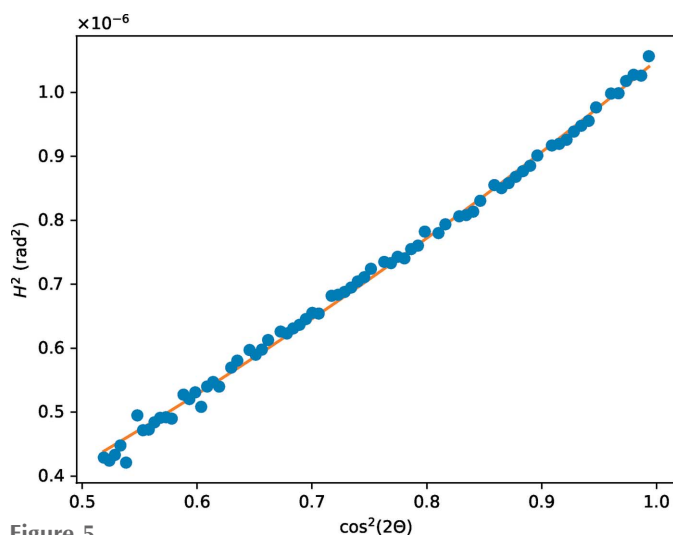


Figure 5
FWHM of diffraction lines for LaB₆ measured at SNBL BM31 at the ESRF using a Mar345 image plate. The solid line shows the resolution function modelled with equation (11).

conventional powder diffraction and PDF measurements. The data were collected using a wavelength of 0.33922 Å (36.55 keV) with the beam focused onto the detector and a sample-to-detector distance of 338 mm. These data were processed the same way as for the previous example.

The data plotted as function of $\cos^2(2\theta)$ (Fig. 5) were fitted with equations (11). For this detector and experimental conditions, the linear B term dominates, therefore not all the parameters can be reliably calculated. Assuming an effective sample size of 0.1 mm, the estimated size of the point spread function p is 0.27 (2) mm, and the effective sensitive layer thickness t is 0.121 (2) mm. All other contributions to the instrumental broadening were found to be negligibly small. The point spread function was expected to be larger than the nominal 0.15 mm pixel size, thus it is expected to be within the range of the two estimates given above.

Therefore, equation (11) does serve as a good model for the experimentally observed broadening and the parameters refined with this model are realistic and meet expectations.

4. A note on the line shape

For powder diffraction, modelling of the diffraction line shape is a necessary component of the Rietveld method and analysis of a microstructure. The above consideration is based on the assumption that the instrumental line L is given by a convolution of n functions modelling different contributions (Scardi, 2020):

$$L = C_1 * C_2 * \dots * C_n. \quad (18)$$

Moreover, we assume that every component can be represented by a normal distribution, which finally gives a Gaussian with the variance

$$\sigma_L^2 = \sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2. \quad (19)$$

If a contribution deviates from Gaussian it has to be taken out from the sum in equation (19) and an exact convolution has to be evaluated for this component. A complete analysis of all components is beyond the scope of the present report; here we only point out one of the origins of the line shift and asymmetry associated with the absorption of the scattered photons in the sensitive layer of a 2D detector. The intensity distribution for an infinitely narrow scattered ray detected by a detector with infinitely small pixels can be written as

$$R(x) = A u(x) \exp\left(-\mu \frac{x}{\sin 2\theta}\right). \quad (20)$$

Here $x = R - R_0$, where R_0 is the radius of the Debye–Scherrer ring on the surface of the detector (see Fig. 6) and R is an apparent radius defined by the photon absorption event in the sensitive layer. A is a normalization coefficient, $u(x)$ is the Heaviside step function defining the thickness profile and μ is the absorption coefficient. The exponential terms represent the absorption and conjugated distribution of probability for a photon to be detected inside the layer, and $x/\sin 2\theta$ is the flight pass of a photon in the layer before the absorption event. Taking the rest of the resolution function as a Gaussian, the total line shape can be written as the convolution

$$L = G * R. \quad (21)$$

A similar convolution was evaluated numerically to model a line asymmetry correction for laboratory diffractometers [see Scardi *et al.* (1994) as an example]. The analytical expression for the convolution of an exponential function and a Gaussian is an exGaussian distribution, and can be written as

$$L(x) = \frac{\lambda}{2} \exp\left(\frac{\lambda}{2}(\lambda\sigma^2 - 2x)\right) \left[1 - \operatorname{erf}\left(\frac{\lambda\sigma^2 - x}{(2)^{1/2}\sigma}\right)\right]. \quad (22)$$

Here $\lambda = \mu/\sin(2\theta)$, and erf is the error function. An illustration given in Fig. 6 shows a specific asymmetry and shift of the maximum relative to the true position. The shift of the line position relates to the mean of equation (22), which is $1/\lambda = \sin(2\theta)/\mu$. The asymmetry may be characterized by a shape factor $K = 1/\sigma\lambda = \sin(2\theta)/\sigma\mu$. Both the shift of line position and the asymmetry of the line shape will increase with the angle of incidence.

Asymmetry is frequently seen for high-angle lines provided that the sample-related broadening is small (see Fig. 6); the shift of the line maxima is more difficult to observe as it is normally redistributed over many lines by the calibration procedure and partially removed by the zero-shift refinement. In order to make it visible, we applied a calibration procedure with *pyFAI* (Kieffer & Karkoulis, 2013) using only low-angle lines in the refinement. This procedure makes the shift for high angles obvious (Fig. 6). However, the above consideration should only be taken as a qualitative illustration and to highlight the need for more detailed quantitative analysis that is still missing for powder diffraction with large 2D area detectors. Here we consider the effect in general; more precise and quantitative analysis can be done on the basis of numerical simulations. For more details on the current state of the development of numerical modelling of the effect and corre-

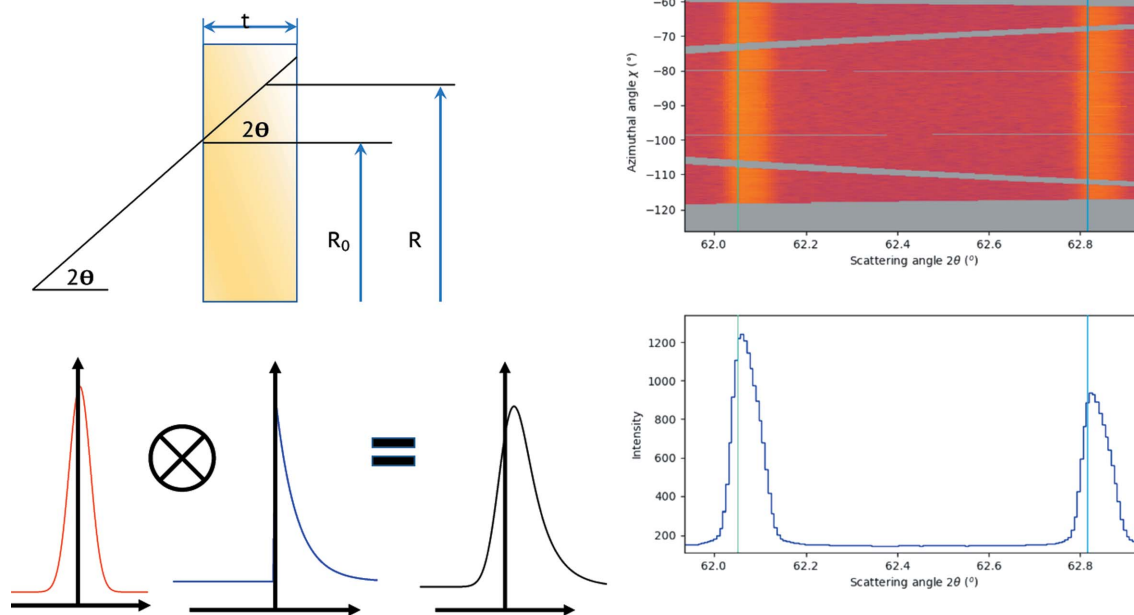


Figure 6

Top left: geometry and notations for equation (20). Bottom left: illustration of convolution equation (21) and the resulting line shape [equation (22)]. Top right: azimuth– 2θ map for LaB_6 pattern measured at a sample-to-detector distance of 138 mm at 19.5 keV. Bottom right: high-angle diffraction lines together with theoretical positions. There is no shift of the line position and no asymmetry for the lines at low angles. Vertical lines show the expected positions of the diffraction peaks. The data were processed by *pyFAI-calib2* (Kieffer *et al.*, 2020).

sponding corrections for the data processing, please see the project by Dr Jerome Kieffer at https://pyfai.readthedocs.io/en/master/usage/tutorial/ThickDetector/Parallax_model.html.

5. Conclusion

Powder diffraction experiments with large area detectors are very common, in particular at synchrotron sources. However, the instrumental broadening of diffraction lines – the resolution function – associated with this experimental geometry is fundamentally different to that observed for the traditional scheme assuming 2θ scanning with point or strip detectors. The illustration is given in Appendix C where FWHMs measured with two different detector schemes are compared.

A new equation [equation (11)] has been derived to model the instrumental resolution function from Bragg's law and the geometry of a diffraction experiment using a 2D area detector together with its corresponding error propagation. Previous efforts to express the instrumental broadening based on geometrical considerations, compared to equation (11), only account for sample size, sample-to-detector distance and point spread function (pixel size), but neglect beam divergence, the thickness of the detector sensitive layer and the uncertainty in the wavelength. Equation (11), therefore, is the most complete model of the instrumental broadening for powder diffraction setups with large 2D detectors.

We have characterized the instrumental resolution using the FWHM of diffraction lines for LaB_6 standard powder neglecting sample-related broadening. The resulting dependencies of the FWHM on the scattering angle fit well to the new model equation (11), with reasonable values of the diffractometer parameters. This conclusion holds for two

different detectors, the Pilatus2M and MAR345, at two different synchrotron beamlines.

The commonly used Caglioti formula for the instrumental broadening corresponds to neither the model equation (11) nor to the experimental observations for this specific geometry. The use of the Caglioti formula in structural refinements thus has the potential danger of incorrect parameterization of the line widths when used in combination with data from large area detectors. Thus, the wrong FWHM stemming from the Caglioti formula may result in incorrect intensities being assigned to overlapping peaks. We therefore derived a replacement of the Caglioti formula with equation (11), have performed experimental verification of the model, and have shown that its parameters are physically meaningful for 2D diffraction geometry.

The new resolution-function model assumes normal distributions for all the resolution components. As a first step beyond this assumption we have considered the effect of absorption in the detector. We have found that use of a thick sensitive detecting layer leads to the asymmetry and shift of the maxima of diffraction lines, and that this effect is more pronounced at high angles. The present analysis is based on a Gaussian line shape for all the resolution components except for the absorption in the detector sensitive layer. A more detailed consideration might be based on more realistic and more frequently used line shapes, such as pseudo-Voigt or Pearson functions. The further development of the model may include analysis of the shape of the sample holder, inhomogeneity of the scattering density of the material inside the sample holder, and the shape and inhomogeneity of the beam.

The resolution function derived here is easy to implement for classical powder diffraction tools such as the Rietveld

method or profile refinements (see Appendix B for a macro for *Topas*). For PDF methods and the case of periodic structures, the new resolution function can be used when the diffraction pattern is modelled in reciprocal space (Neder & Proffen, 2020). Equation (11) can also be easily rewritten as a function of Q , since $\cos 2\theta = 1 - 2 \sin^2 \theta = 1 - (\lambda/2\pi)Q$. The resulting polynomial form obviously differs from the frequently assumed ' $\sigma_0^2 + \alpha^2 Q^2$ ' dependence (Qiu *et al.*, 2004). The corresponding corrections to be applied for calculations of the radial pair-distribution function $G(r)$ have still to be derived.

APPENDIX A

Derivations and illustrations of the resolution function

For the sake of clarification, we provide here some details of the derivations omitted in the main text.

Evaluation of contributions on the left-hand side of equation (6):

$$\frac{\partial d}{\partial D} \delta D = \frac{\lambda}{4D} \frac{\cos \theta}{\sin^2 \theta} \frac{RD}{R^2 + D^2} \delta D = \frac{1}{2} \frac{\delta D}{D} d \cot \theta \sin 2\theta \cos 2\theta, \quad (23)$$

$$\frac{\partial d}{\partial R} \delta R = \frac{\lambda}{4R} \frac{\cos \theta}{\sin^2 \theta} \frac{RD}{R^2 + D^2} \delta R = \frac{1}{2} \frac{\delta R}{R} d \cot \theta \sin 2\theta \cos 2\theta, \quad (24)$$

$$\frac{\partial d}{\partial \lambda} \delta \lambda = \frac{\delta \lambda}{2 \sin \theta} = \frac{\delta \lambda}{\lambda} d. \quad (25)$$

These contributions squared and summed up give equation (7).

Estimates of longitudinal and transverse uncertainties were obtained as follows. A photon can be scattered somewhere inside the sample with lateral size c and then detected inside the sensitive layer with an effective thickness t ; these scattering and absorption processes define the uncertainty in the sample-to-detector distance. Spatial distributions of the probabilities of these events may be quite complex. Here we assume a normal distribution and take half of the maximal deviation from the ideal geometry (point sample and infinitely thin detecting layer) as corresponding variances. This estimate can be taken as an upper limit, as the corresponding distributions are broader than the physical sizes of sample and sensitive layer. Error propagation gives

$$(\delta D)^2 \approx \sigma_D^2 \approx \frac{1}{4}(c^2 + t^2). \quad (26)$$

Similarly for the transverse component, one needs to account for the probability of a photon to be scattered with angle 2θ in a sample with transverse size c (we explicitly consider only the symmetric case when the lateral and transverse dimensions of the sample and the beam are the same), which is then detected along the transverse component of the flight path in the detecting layer, $t \tan 2\theta$, in the detector with a point spread function or pixel size p . One also needs to account for the

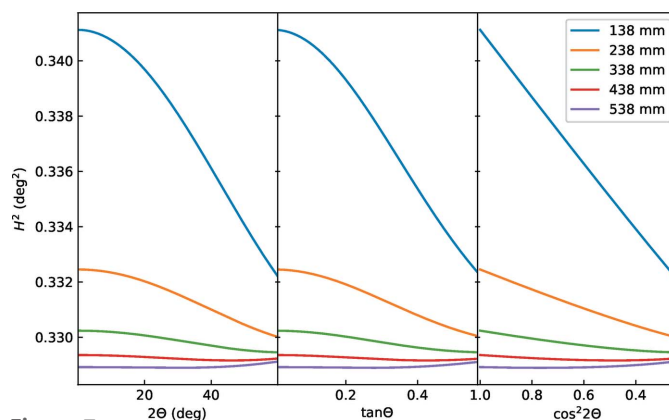


Figure 7
Resolution function for various sample-to-detector distances.

width of the scattered ray (divergence φ), which contributes to the transverse smearing as $\varphi D / \cos^2 2\theta$ (Fig. 2). Again assuming normal distributions for the scattering and absorption processes and taking half of the maximal deviation from the ideal case as the corresponding variance, the transverse variance reads

$$(\delta R)^2 \approx \sigma_R^2 \approx \frac{1}{4} \left(\frac{c^2}{\cos^2 2\theta} + p^2 + t^2 \tan^2 2\theta + \frac{\varphi^2 D^2}{\cos^4 2\theta} \right). \quad (27)$$

Equation (10) can be rewritten in the following form:

$$(\delta \theta)^2 = \frac{1}{16D^2} \left[\cos^4 2\theta (p^2 - 2t^2 - c^2) + \cos^2 2\theta (2c^2 + 2t^2) + D^2 \varphi^2 \right]. \quad (28)$$

In order to illustrate the effect of various parameters of the experiment, we have simulated the instrumental broadening of the diffraction lines according to the theory presented above. For convenience, every simulation is given with the instrumental FWHM as a function of the Bragg angle 2θ , as a function of $\tan \theta$ for inspection of applicability of the Caglioti formula, and as a function of $\cos^2(2\theta)$.

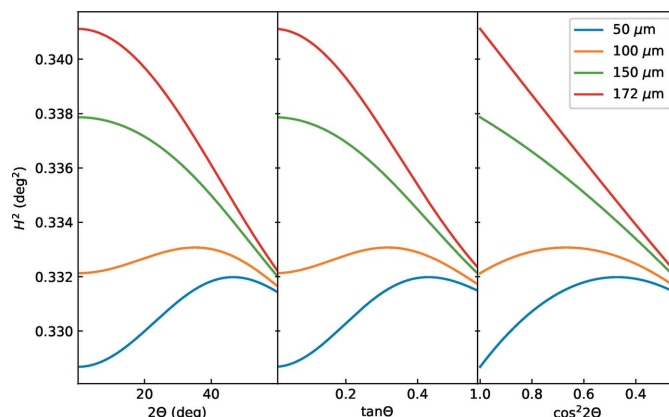


Figure 8
Resolution function for various pixel sizes.

A1. Effect of sample-to-detector distance on the instrumental FWHM

See Fig. 7, for which all the other parameters from equation (11) are fixed at the following values:

- pixel size $p = 172 \mu\text{m}$;
- effective sensor thickness $t = 100 \mu\text{m}$;
- capillary size $c = 100 \mu\text{m}$;
- beam divergence $\varphi = 0.01 \text{ mrad}$;
- wavelength $\lambda = 0.7 \text{ \AA}$; and
- wavelength uncertainty $\delta\lambda = 5 \times 10^{-8} \text{ \AA}$.

A2. Effect of pixel size on the instrumental FWHM

See Fig. 8, for which all the other parameters from equation (11) are fixed at the following values:

- sample-to-detector distance $D = 138 \text{ mm}$;
- effective sensor thickness $t = 100 \mu\text{m}$;
- capillary size $c = 100 \mu\text{m}$;
- beam divergence $\varphi = 0.01 \text{ mrad}$;
- wavelength $\lambda = 0.7 \text{ \AA}$; and
- wavelength uncertainty $\delta\lambda = 5 \times 10^{-8} \text{ \AA}$.

A3. Effect of sensitive-layer thickness on the instrumental FWHM

See Fig. 9, for which all the other parameters from equation (11) are fixed at the following values:

- sample-to-detector distance $D = 138 \text{ mm}$;
- pixel size $p = 172 \mu\text{m}$;
- capillary size $c = 100 \mu\text{m}$;
- beam divergence $\varphi = 0.01 \text{ mrad}$;
- wavelength $\lambda = 0.7 \text{ \AA}$; and
- wavelength uncertainty $\delta\lambda = 5 \times 10^{-8} \text{ \AA}$.

A4. Effect of capillary size on the instrumental FWHM

See Fig. 10, for which all the other parameters from equation (11) are fixed at the following values:

- sample-to-detector distance $D = 138 \text{ mm}$;
- pixel size $p = 172 \mu\text{m}$;
- effective sensor thickness $t = 100 \mu\text{m}$;
- beam divergence $\varphi = 0.01 \text{ mrad}$;
- wavelength $\lambda = 0.7 \text{ \AA}$; and
- wavelength uncertainty $\delta\lambda = 5 \times 10^{-8} \text{ \AA}$.

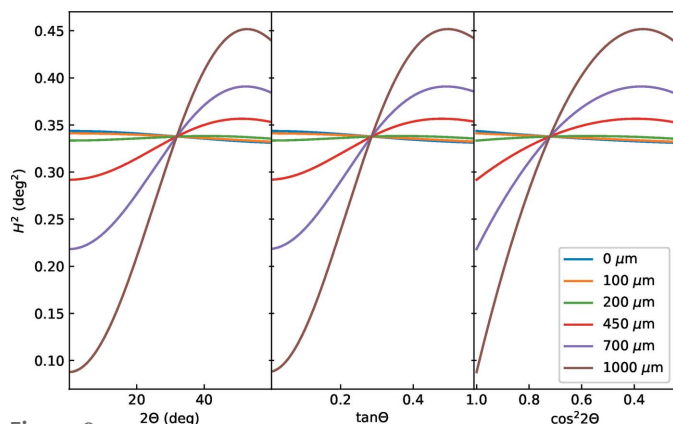


Figure 9
Resolution function for various sensitive-layer thicknesses.

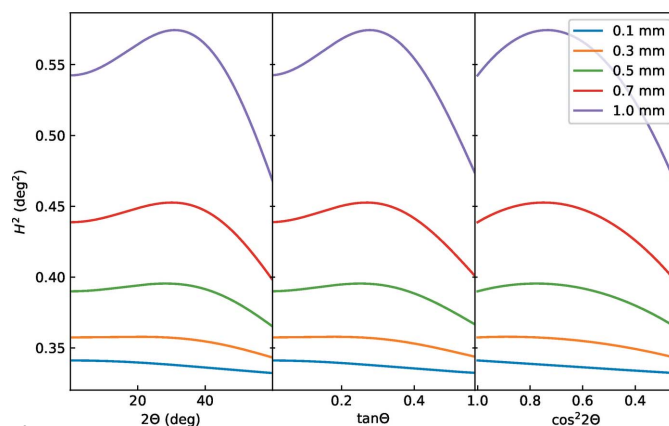


Figure 10
Resolution function for various capillary sizes.

APPENDIX B

Topas macro

```
macro DC1(a, av, b, bv, c, cv)
{
  #m_argu a
  #m_argu b
  #m_argu c
  If_Prm_Eqn_Rpt(a, av, min -1 max 2)
  If_Prm_Eqn_Rpt(b, bv, min -1 max 1)
  If_Prm_Eqn_Rpt(c, cv, min -1 max 2)
  gauss_fwhm = sqrt(CeV(a, av) Cos(2 * Th)^4
    + CeV(b, bv) Cos(2 * Th)^2 + CeV(c, cv));
}
```

Here the parameters a, b, c are the coefficients A, B, C from equation (11). Their starting values can be estimated with help of equation (12) if the sample-to-detector distance, sample

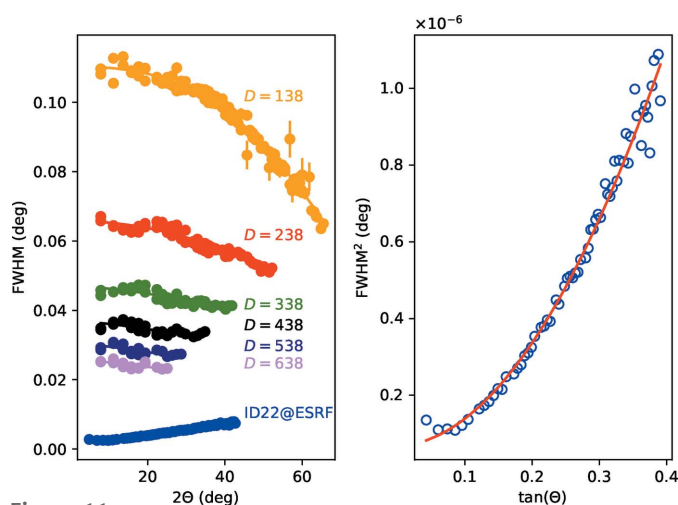


Figure 11
Comparison of the FWHM for various distances with LaB_6 measured at the SNBL BM01@ESRF beamline and the ID22@ESRF high-resolution powder diffraction beamline. Note the opposite trends for the FWHM as a function of scattering angle. For the ID22@ESRF scanning 2θ diffractometer the line broadening corresponds to the Caglioti function (right panel), while for SNBL BM01@ESRF with a 2D detector the data are parameterized with equation (11) (left panel).

size, pixel size, effective thickness and beam divergence are known.

APPENDIX C

Resolution function for diffractometers with a 2D detector and multi-analyser detector stage

The resolution function for this case is shown in Fig. 11.

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References

- Ashiotis, G., Deschildre, A., Nawaz, Z., Wright, J. P., Karkoulis, D., Picca, F. E. & Kieffer, J. (2015). *J. Appl. Cryst.* **48**, 510–519.
- Caglioti, G., Paoletti, A. & Ricci, F. (1958). *Nucl. Instrum.* **3**, 223–228.
- Dyadkin, V., Pattison, P., Dmitriev, V. & Chernyshov, D. (2016). *J. Synchrotron Rad.* **23**, 825–829.
- Gozzo, F., De Caro, L., Giannini, C., Guagliardi, A., Schmitt, B. & Prodi, A. (2006). *J. Appl. Cryst.* **39**, 347–357.
- Hinrichsen, B., Dinnebier, R. E. & Jansen, M. (2008). *Powder Diffraction: Theory and Practice*, pp. 414–438. London: The Royal Society of Chemistry. <http://dx.doi.org/10.1039/9781847558237-00414>.
- Kieffer, J. & Karkoulis, D. (2013). *J. Phys. Conf. Ser.* **425**, 202012.
- Kieffer, J., Valls, V., Blanc, N. & Hennig, C. (2020). *J. Synchrotron Rad.* **27**, 558–566.
- Klug, H. P. & Alexander, L. E. (1974). *X-ray Diffraction Procedures: for Polycrystalline and Amorphous Materials*. New York: Wiley. <http://lib.ugent.be/catalog/rug01:000473316>.
- Neder, R. B. & Proffen, Th. (2020). *J. Appl. Cryst.* **53**, 710–721.
- Norby, P. (1997). *J. Appl. Cryst.* **30**, 21–30.
- Pecharsky, V. K. & Zavalij, P. Y. (2009). *Fundamentals of Powder Diffraction and Structural Characterization of Materials*. Boston: Springer. <http://dx.doi.org/10.1007/978-0-387-09579-0>.
- Qiu, X., Božin, E. S., Juhas, P., Proffen, T. & Billinge, S. J. L. (2004). *J. Appl. Cryst.* **37**, 110–116.
- Scardi, P. (2020). *Cryst. Growth Des.* **20**, 6903–6916.
- Scardi, P., Lutterotti, L. & Maistrelli, P. (1994). *Powder Diffr.* **9**, 180–186.